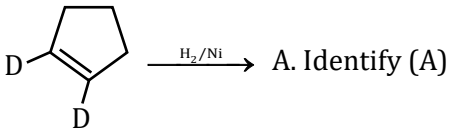
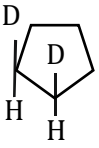
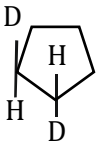
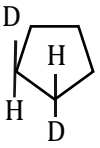
EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Preparation of alkane by hydrogenation of alkene

1. 

A. Identify (A)

(A) $CH_3-(CH_2)_4-CH_3$ (B)  (C)  (D) 

CHC087

2. Which of the following compound cannot be produced by hydrogenation of alkene :-

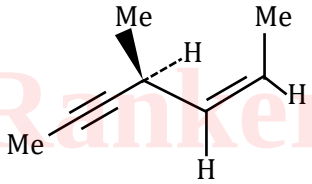
(A) Propane (B) Isobutane (C) Neopentane (D) Iso-octane

CHC088

3. $CH_3—C\equiv CH + 2H_2 \xrightarrow{Pt}$ P (major). Identify (P)

(A) $CH_3CH = CH_2$ (B) $CH_3CH_2CH_3$ (C) Both A and B (D) None of these

CHC089

4. 

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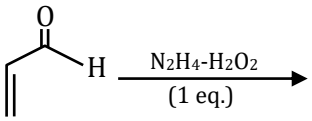
Hydrogenation of the above compound in the presence of poisoned paladium catalyst gives –

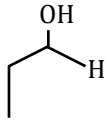
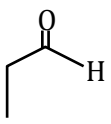
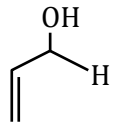
(A) An optically active compound (B) An optically inactive compound

(C) A racemic mixture (D) A diastereomeric mixture

CHC053

Preparation of alkane by Transfer hydrogenation (NH_2-NH_2+H_2O_2)

5. 

(A)  (B)  (C)  (D) All of these

CHC091

Preparation of alkane by decarboxylation (Soda-lime)

6. The number of isomeric monocarboxylic acids which on decarboxylation form isopentane is :

(A) 2 (B) 1 (C) 3 (D) 4

CHC103

7. By wurtz reaction, a mixture of methyl iodide and ethyl iodide gives -

(A) Propane (B) Ethane

(C) butane (D) a mixture of the above three

CHC093

For More Join: @IITJEE_Advanced

8. $C_2H_5Br + CH_3-\underset{\underset{CH_3}{|}}{CH}-Br \xrightarrow{Na/\text{dry ether}}$ 'P ' (major)

Which is not product 'P' –

(A) $CH_3—CH_2—CH_2—CH_3$

(B) $CH_3—CH_2—\underset{\underset{CH_3}{|}}{CH}—CH_3$

(C) $CH_3—\underset{\underset{CH_3}{|}}{CH}—\underset{\underset{CH_3}{|}}{CH}—CH_3$

(D) $CH_3—CH_2—CH_2—CH_2—CH_3$

CHC094

9. $CH_3CH_2CH_2Cl \xrightarrow{Zn/H^{\oplus}} (P)$

Product (P) of the above reaction is : -

(A) $CH_3CH_2CH_2CH_2CH_2CH_3$

(B) $CH_3CH = CH_2$

(C) $CH_3CH_2CH_3$

(D) Both (i) and (iii)

CHC096

10. The order of reactivity of alkyl halides in Wurtz reaction is

(A) R–I > R–Br > R–Cl

(B) R–I < R–Br < R–Cl

(C) R–Br > R–I < R–Cl

(D) R–I > R–Cl > R–Br

CHC092

11. If isopropyl chloride and propyl chloride both react with Na in presence of dry ether which alkane obtains :-

(a) $CH_3—\underset{\underset{CH_3}{|}}{CH}—CH_2—CH_2—CH_3$

(b) $CH_3—\underset{\underset{CH_3}{|}}{CH}—CH_2—CH_3$

(c) $CH_3—\underset{\underset{CH_3}{|}}{CH}—\underset{\underset{CH_3}{|}}{CH}—CH_3$

(d) $CH_3—CH_2—CH_2—CH_2—CH_2—CH_3$

(A) a, b only (B) a, c, d only (C) b, c, d only (D) All of these

CHC095

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Preparation of alkane by Kolbe's electrolysis

12. During the preparation of ethane by Kolbe's electrolytic method using inert electrodes the pH of the electrolyte –

(A) Increases progressively as the reaction proceeds

(B) Decreases progressively as the reaction proceeds

(C) Remains constant throughout the reaction

(D) May decrease of the concentration of the electrolyte is not very high

CHC101

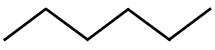
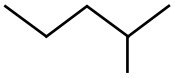
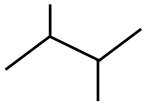
13. Kolbe electrolytic synthesis can be used to obtain all hydrocarbons as a major product except :-

(A) 2-Butene (B) Ethyne (C) Methane (D) Ethene

CHC216

Physical properties and uses of alkane

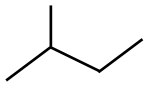
14. Which of the following has the lowest boiling point?

(A)  (B)  (C)  (D) 

CHC108

For More Join: @IITJEE_Advanced

15. Arrange the following alkanes in decreasing order of their heat of combustion.

(i)  (ii)  (iii) 

(A) (i) > (ii) > (iii) (B) (iii) > (i) > (ii) (C) (iii) > (ii) > (i) (D) (i) > (iii) > (ii)

CHC109

16. Arrange the following in the decreasing order of their melting points:

(i) Decane (ii) Nonane (iii) Octane (iv) Heptane

(A) (i) > (ii) > (iii) > (iv) (B) (iv) > (iii) > (ii) > (i)

(C) (i) > (iii) > (ii) > (iv) (D) (iv) > (ii) > (iii) > (i)

CHC111

17. Hydrocarbon that is liquid at room temperature is:

(A) Ethane (B) Propane (C) Butane (D) Pentane

CHC112

Chemical reaction of alkane; Halogenation (Free radical substitution), Reactivity and selectivity principle, Relative rate of reaction (X_2, Alkane), Reed Reaction

18. Photochemical chlorination of alkane is initiated by a process of

(A) pyrolysis (B) substitution (C) Homolysis (D) Peroxidation

CHC115

19. The reactivity order of halogens during halogenation of alkane decreases in the order -

(A) $F_2 > Cl_2 > Br_2 > I_2$ (B) $I_2 > Br_2 > Cl_2 > F_2$ (C) $F_2 > Cl_2 > I_2 > Br_2$ (D) $Cl_2 > F_2 > Br_2 > I_2$

CHC117

20. Reaction $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ follows free radical mechanism. The chain termination step would not include the process :-

(A) $\dot{C}H_3 + \dot{C}H_3 \longrightarrow CH_3 - CH_3$ (B) $\dot{C}H_3 + \dot{C}l \longrightarrow CH_3 - Cl$

(C) $\dot{H} + \dot{C}l \longrightarrow HCl$ (D) $\dot{C}l + \dot{C}l \longrightarrow Cl_2$

CHC119

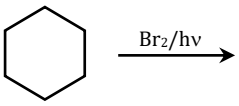
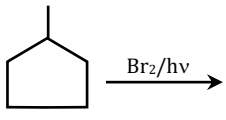
21. $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_3 \xrightarrow[h\nu]{Cl_2}$

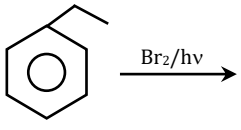
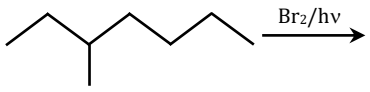
Number of chiral centers generated during monochlorination in the above reaction :

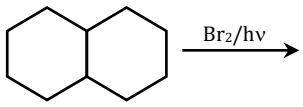
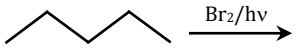
(A) 1 (B) 2 (C) 3 (D) 4

CHC019

22. Among the following free radical bromination reactions, select those in which 2° halide is the major product-

(A)  (B) 

(C)  (D) 

(E)  (F) 

(A) A, B, C, D (B) A, C, F (C) A, C, D, E (D) A, B, C, D, E

CHC056

For More Join: @IITJEE_Advanced

Reactions of alkanes; combustion; controlled oxidation [$KMnO_4$, Cu/Heat, Mo_2O_3, $(CH_3COO)_2Mn$]; isomerization; pyrolysis; aromatization and reaction with steam

23. A mixture of propene and methane is obtained by the cracking of :-

(A) 1-butene (B) 2-butene (C) n- butane (D) Isobutane

CHC120

24. n-hexane $\xrightarrow[600°C]{Cr_2O_3/Al_2O_3}$ Cyclic compound. The wrong title of the reaction would be :-

(A) Aromatization (B) Hydroformation (C) Cyclisation (D) Isomerisation

CHC122

25. When n-butane is heated in the presence of $AlCl_3/HCl$ it will be converted into –

(A) Ethane (B) Propane (C) Butene (D) Isobutane

CHC124

26. Pyrolysis of alkanes is a

(A) Nucleophilic addition reaction (B) Free radical substitution reaction

(C) Electrophilic addition reaction (D) Free radical elimination reaction

CHC126

27. Which of the following reactions of methane is complete combustion :-

(A) $2CH_4 + O_2 \xrightarrow{Cu/523K/100atm.} 2CH_3OH$ (B) $CH_4 + O_2 \xrightarrow[\Delta]{Mo_2O_3} HCHO + H_2O$

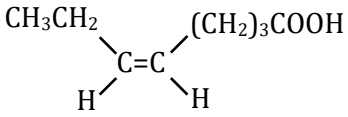
(C) $CH_4 + O_2 \longrightarrow C(s) + 2H_2O(\ell)$ (D) $CH_4 + 2O_2 \longrightarrow CO_2(g) + 2H_2O(\ell)$

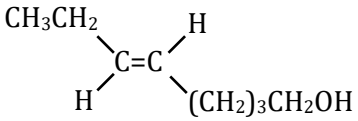
CHC127

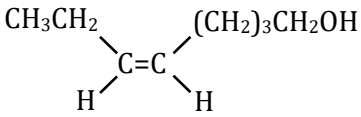
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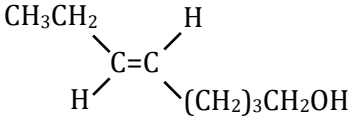
Preparation of alkene by lindlar's catalyst, P-2 catalyst (Ni_2B/H_2), poisoned catalyst

28. $CH_3CH_2–C\equiv C–(CH_2)_3–COOH \xrightarrow[S\text{-quinoline}]{H_2,Pd\text{-}BaSO_4}$ P (Major Product). Identify P

(A) 

(B) 

(C) 

(D) 

CHC131

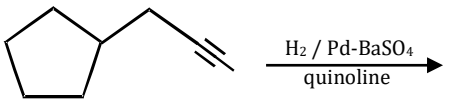
29. When 2- butyne is treated with H_2, $Pd–BaSO_4$; the product formed will be :

(A) Cis –2– butene (B) Trans –2– butene

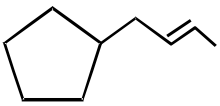
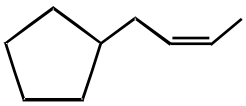
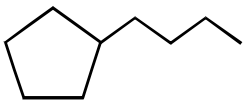
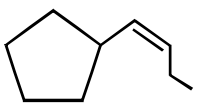
(C) 1 – butene (D) 2– hydroxy butane

CHC229

30. The product formed in the reaction

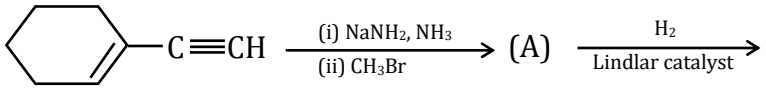


is :

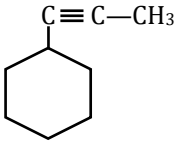
(A)  (B)  (C)  (D) 

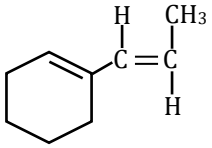
CHC230

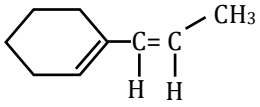
For More Join: @IITJEE_Advanced

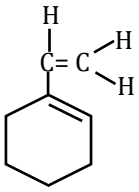
31. 

(B) ; Product (B) is :

(A) 

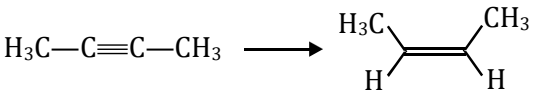
(B) 

(C) 

(D) 

CHC016

32. The most suitable reagent for the following conversion is :

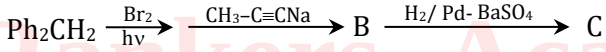


cis-2-butene

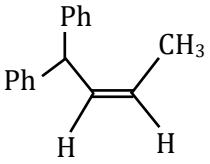
(A) Na/liquid NH_3 (B) H_2, Pd/C, quinoline

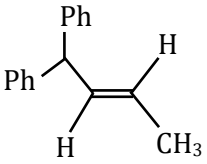
(C) Zn/HCl (D) $Hg^{2\oplus}/H^{\oplus}$, H_2O

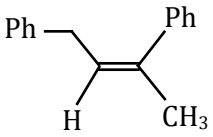
CHC223

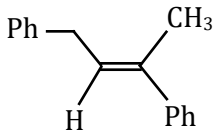
33. 

The product (C) is:

(A) 

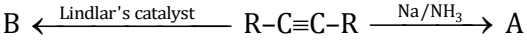
(B) 

(C) 

(D) 

CHC233

Preparation of alkene by birch reduction

34. 

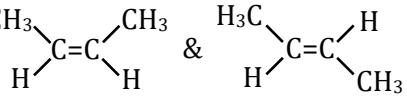
A and B are geometrical isomers –

(A) A is trans, B is cis (B) A and B both are cis

(C) A and B both are trans (D) A is cis, B is trans

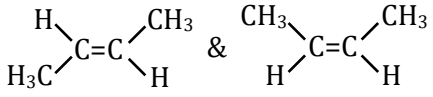
CHC060

35. When a gas (x) is passed through red hot iron tube benzene is obtained. (x) on treatment with $H_2/Pd\text{-}BaSO_4$ & $Na/liq.NH_3$ gives respectively :-

(A) 

(B) $^{\oplus}Na\,^{\ominus}C{\equiv}C^{\ominus}Na^{\oplus}$ & $CH_2{=}CH_2$

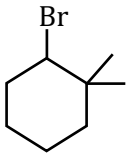
(C) $CH_2{=}CH_2$ & $^{\oplus}Na^{\ominus}C{\equiv}C^{\ominus}Na^{\oplus}$

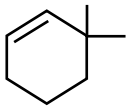
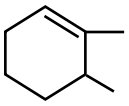
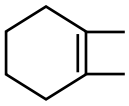
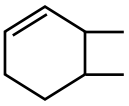
(D) 

CHC231

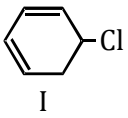
For More Join: @IITJEE_Advanced

Basic idea of Elimination reactions

36.  $\xrightarrow[\Delta]{\text{alcoholic KOH}}$ Product ; Product is:

(A)  (B)  (C)  (D) 

CHC133

37. 

I II III

Which one is most easily dehydrohalogenated?

(A) I (B) II (C) III (D) Both A and B

CHC135

38. $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-CH_2CH_2CH_2OH \xrightarrow{H^{\oplus}}$ Alkene (Major product)

The major product formed in the above reaction will be:-

(A) $H_3C-\underset{\displaystyle H_3C}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}=\underset{\displaystyle CH_2-CH_3}{C}$

(B) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-CH=CH-CH_3$

(C) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-CH_2-CH=CH_2$

(D) None of these

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CHC136

39. The order of reactivity of alcohols with respect to acid catalysed elimination will be :

(a) $(CH_3)_3C-OH$ (b) $CH_3CH_2CH_2CH_2OH$ (c) $CH_3\underset{\displaystyle OH}{\overset{|}{\underset{|}{C}}}H-CH_2-CH_3$

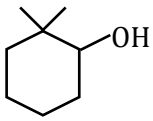
(A) a > b > c (B) a > c > b (C) b > a > c (D) b > c > a

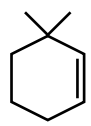
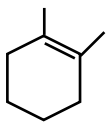
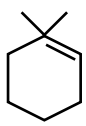
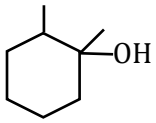
CHC137

40. Sec. butyl alcohol on dehydration gives the predominant product :-

(A) 2-Butene (B) 1-Butene (C) Isobutene (D) All of the these

CHC138

41.  $\xrightarrow{H^{\oplus}}$ P. The product P is :

(A)  (B)  (C)  (D) 

CHC139

For More Join: @IITJEE_Advanced

42. Which will dehydrate at fastest rate by H_3PO_4 :

(A) 2-methyl butan-2-ol (B) 3-methyl butan-2-ol

(C) Butan-1-ol (D) 2-methyl butan-1-ol

CHC144

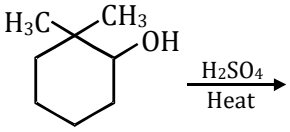
43. The reaction of tert. butyl bromide with sodium methoxide gives :-

(A) isobutane (B) tert. butylmethyl ether

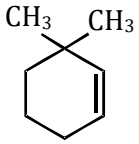
(C) isobutylene (D) sodium tert. Butoxide

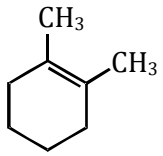
CHC145

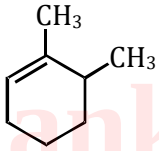
44. Consider the reaction

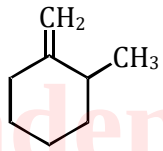


The alkene formed in major amount is

(A) 

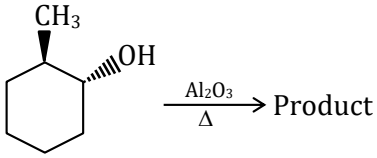
(B) 

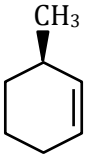
(C) 

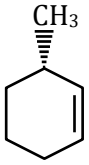
(D) 

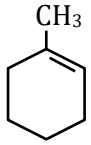
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CHC146

45. 

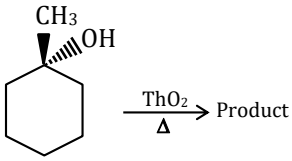
(A) 

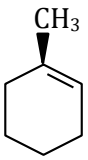
(B) 

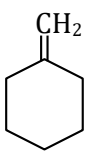
(C) 

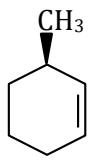
(D) Both (A) & (B)

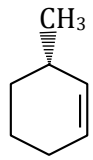
CHC150

46. 

(A) 

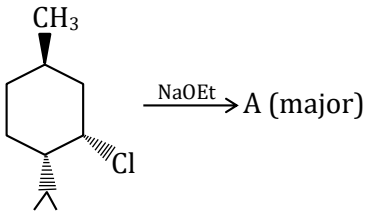
(B) 

(C) 

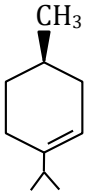
(D) 

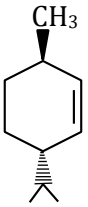
CHC151

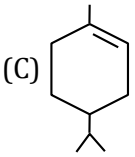
For More Join: @IITJEE_Advanced

47. 

Identify product 'A'

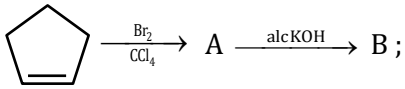
(A) 

(B) 

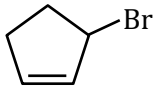
(C) 

(D) None of these

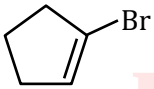
CHC155

48. 

What is the structure of B

(A) 

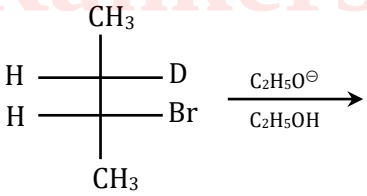
(B) 

(C) 

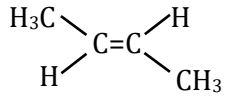
(D) 

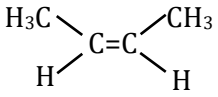
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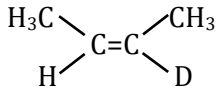
CHC167

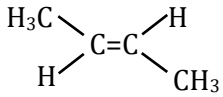
49. 

The Alkene formed is mainly :

(A) 

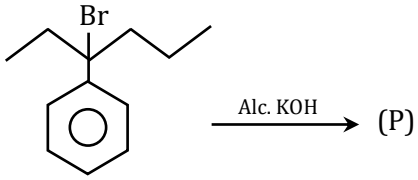
(B) 

(C) 

(D) 

CHC007

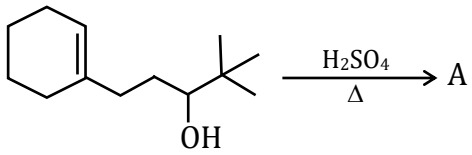
50. How many total dehydrohalogenation products are formed in following reaction.



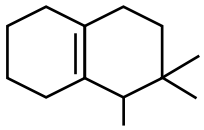
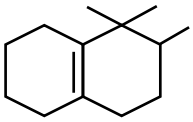
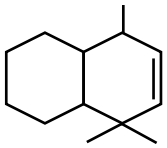
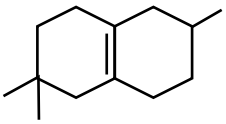
(A) 3 (B) 4 (C) 5 (D) 6

CHC008

For More Join: @IITJEE_Advanced

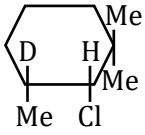
51. 

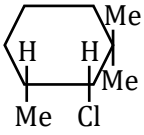
Product (A) is :

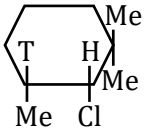
(A)  (B)  (C)  (D) 

CHC015

52. Identify correct reactivity order for E_2 reaction with alcoholic KOH

(i) 

(ii) 

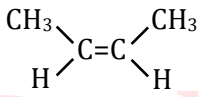
(iii) 

(A) (i) > (ii) > (iii) (B) (ii) > (i) > (iii) (C) (i) > (iii) > (ii) (D) (iii) > (i) > (ii)

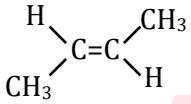
CHC057

Physical properties and uses of alkene

53. Which one is correct about the boiling point of Cis-But-2-ene and trans-But-2-ene :-



(A)



(B)

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(A) (A) > (B) (B) (B) > (A) (C) (A) = (B) (D) Cannot compare

CHC156

54. Out of o-xylenes, m-xylenes, p-xylenes, which have highest melting point.

(A) o-xylenes (B)m-xylenes (C) p-xylenes (D) All have same M.P.

CHC157

55. Which one is the correct order of solubility in non-polar solvents :-

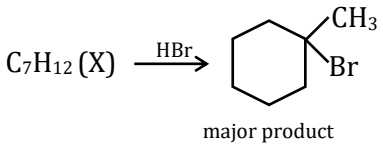
(A) alkane > alkynes > alkene (B) alkane > alkene > alkynes

(C) alkenes > alkynes > alkane (D) alkynes > alkenes > alkane

CHC158

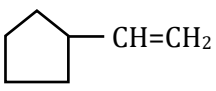
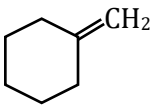
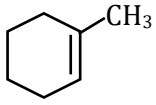
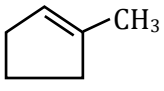
Chemical reaction of alkene; basic idea of addition reaction

56. In the given reaction



major product

(X) cannot be :

(A)  (B)  (C)  (D) 

CHC018

For More Join: @IITJEE_Advanced

57. Anti–Markovnikov addition of HBr is not observed in –

(A) Propene (B) But–2–ene (C) But–1–ene (D) Pent–2–ene

CHC160

58. Statement-1: HCl do not show peroxide effect.

Statement-2: HCl bond is strong enough to break by peroxide in presence of alkene, so it prefers to heterolysis than homolytic.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

CHC260

59. The addition of HCl to 3, 3, 3–trichloropropene gives

(A) $Cl_3CCH_2CH_2Cl$ (B) $Cl_3CCH(Cl)CH_3$ (C) $Cl_2CHCH(Cl)CH_2Cl$ (D) $Cl_2CHCH_2CHCl_2$

CHC161

60. 2,3–Dimethyl–2–butene can be prepared by heating which of the following compounds with a strong acid?

(A) $(CH_3)_2C{=}CH{-}CH_2{-}CH_3$

(B) $(CH_3)_2CH{-}CH_2{-}CH{=}CH_2$

(C) $(CH_3)_2CH{-}\underset{\displaystyle CH_3}{\underset{|}{CH}}{-}CH{=}CH_2$

(D) $(CH_3)_3C{-}CH{=}CH_2$

CHC163

61. $CH_3CH{=}CH$–C$_6$H$_4$–OCH_3 on reaction with HBr gives :-

(A) $CH_3\underset{\displaystyle Br}{\underset{|}{CH}}\underset{\displaystyle H}{\underset{|}{CH}}$–C$_6H_4$–$OCH_3$

(B) $CH_3\underset{\displaystyle H}{\underset{|}{CH}}\underset{\displaystyle Br}{\underset{|}{CH}}$–C$_6H_4$–Br

(C) $CH_3\underset{\displaystyle H}{\underset{|}{CH}}\underset{\displaystyle Br}{\underset{|}{CH}}$–C$_6H_4$–$OCH_3$

(D) $CH_3\underset{\displaystyle Br}{\underset{|}{CH}}$–C$_6H_4$–Br

CHC165

62. $CH_3 - CH = CH_2 \xrightarrow[CCl_4]{Br_2}$ Product

True statement is :-

(A) Reaction completes with anti addition mechanism.

(B) Reaction can be used as a test for unsaturation

(C) Product is optically inactive due to external compensation

(D) All above these

CHC169

63. $CH_3 - CH = CH_2 + NOCl \rightarrow P$. Identify the adduct:-

(A) $CH_3\underset{\displaystyle Cl}{\underset{|}{CH}} - \underset{\displaystyle NO}{\underset{|}{CH_2}}$

(B) $CH_3\underset{\displaystyle NO}{\underset{|}{CH}} - \underset{\displaystyle Cl}{\underset{|}{CH_2}}$

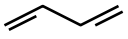
(C) $CH_3{-}CH_2 - \underset{\displaystyle Cl}{\underset{|}{\overset{\displaystyle NO}{\overset{|}{CH}}}}$

(D) $\underset{\displaystyle NO}{\underset{|}{CH_2}} - CH_2 - \underset{\displaystyle Cl}{\underset{|}{CH_2}}$

CHC171

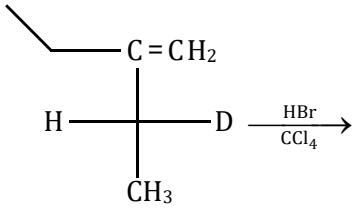
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64. Which of the following will be most reactive in the addition reaction with HBr?

(A) $CH_3 - C \equiv C - CH_3$ (B) 

(C)  (D) None of these

CHC190

65. 

Product (without rearrangement of carbocation)

What is the stereochemistry of product formed :

(A) Racemic mixture (B) Optically inactive

(C) Mixture of diastereomers (D) Meso product

CHC001

66. The reaction of propene with HCl proceeds through the intermediate

(A) $CH_3 - \overset{\oplus}{C}H - CH_3$ (B) $CH_3 - CH_2 - \overset{\oplus}{C}H_2$

(C) $CH_3 - CHCl - \overset{\oplus}{C}H_2$ (D) $CH_3 - \overset{\oplus}{C}H - CH_2 - OH$

CHC012

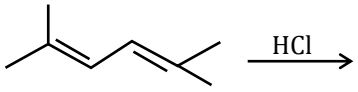
67. In which of the following reaction, Markovnikov rule is violated?

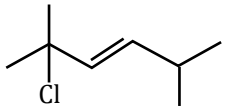
(A) $CH_3-O-CH=CH_2 \xrightarrow[CCl_4]{HBr}$ (B) $CH_3-NH-CH=CH_2 \xrightarrow[CCl_4]{HBr}$

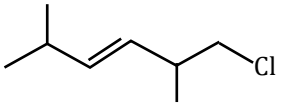
(C) $CH_3-S-CH=CH_2 \xrightarrow[CCl_4]{HBr}$ (D) $O_2N-CH=CH_2 \xrightarrow[CCl_4]{HBr}$

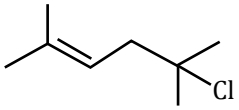
CHC013

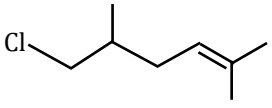
68. What is the product of 1, 4-addition in the reaction shown below?



(A) 

(B) 

(C) 

(D) 

CHC014

Ozonolysis of alkenes (Reductive and oxidative)

69. Reductive ozonolysis of 3–Methyl–1–butene gives a mixture of

(A) Propanal and ethanal (B) Propanone and ethanal

(C) 2–Methylpropanal and methanal (D) Butanone and methanal

CHC180

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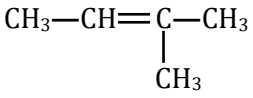
70. Which of the compounds with molecular formula C_5H_{10} yields acetone on reductive ozonolysis
(A) 3-Methyl-1-butene (B) Cyclopentene (C) 2-Methyl-1-butene (D) 2-Methyl-2-butene

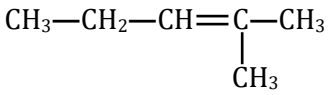
CHC185

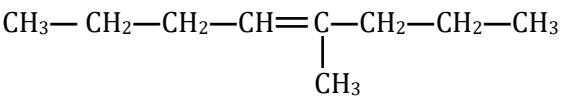
71. Reductive ozonolysis of acetylene gives-
(A) Oxalic acid (B) Ethylene glycol (C) Glyoxal (D) CH_3CHO

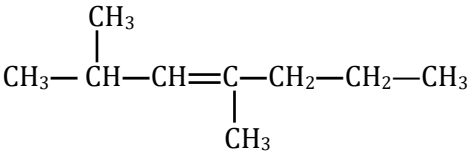
CHC189

72. Hydrocarbon 'A' on reductive ozonolysis gives two isomeric forms 'B' and 'C'. B on oxidation using tollen's reagent gives 'D'. Structure of hydrocarbon is

(A) 

(B) 

(C) 

(D) 

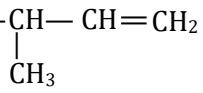
CHC005

Oxidation of Alkenes (Baeyer's Reagent, OsO_4, Hot $KMnO_4$, Acidic $KMnO_4$, Peroxy Acid followed by hydrolysis, Ag/O_2 followed by hydrolysis)

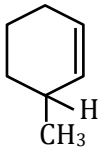
73. The treatment of $(CH_3)_2CCHCH_3$ with boiling $KMnO_4$ produces
(A) $CH_3COCH_3 + CH_3COOH$ (B) $CH_3COCH_3 + CH_3CHO$
(C) $CH_3CHO + CO_2$ (D) CH_3COCH_3 only

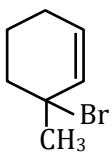
CHC199

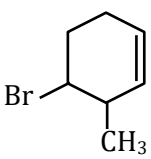
Allylic substitution (NBS/Br_2)

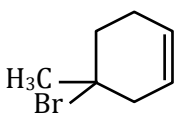
74. Which of the following is maximum reactive for NBS ?
(A) $PhCH_3$ (B) $PhCH_2—CH_3$ (C) $PhCH_2—CH=CH_2$ (D) 

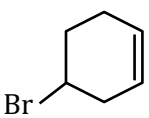
CHC184

75.  major product will be :-

(A) 

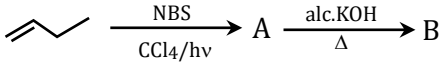
(B) 

(C) 

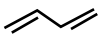
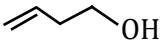
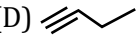
(D) 

CHC186

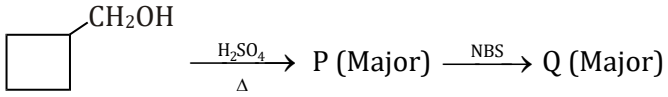
For More Join: @IITJEE_Advanced

76. 

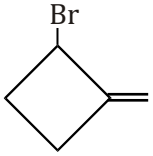
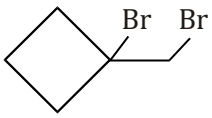
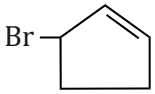
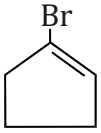
Identify 'B' product :-

(A)  (B)  (C)  (D) 

CHC188

77. 

The structure of Q is

(A)  (B)  (C)  (D) 

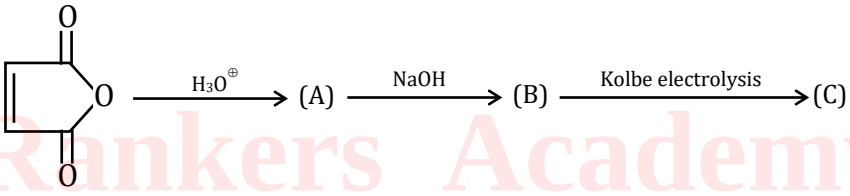
CHC059

Preparations of Alkynes by Kolbe's Electrolysis

78. Kolbe's electrolysis of sodium maleate at anode gives.

(A) Only acetylene (B) Only CO_2 (C) Acetylene + CO_2 (D) CO_2 + H_2

CHC217

79. 

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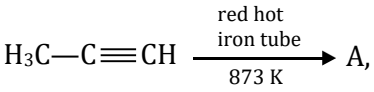
Product (C) of above reaction is:

(A) $H_2C = CH_2$ (B) $CH_3 - C \equiv C - CH_3$ (C) $HC \equiv CH$ (D) $CH_3 - CH = CH - CH_3$

CHC017

Chemical reaction of alkyne ; Dimer, Trimer, Tetramer of acetylene

80. In the following reaction,



the number of sigma (σ) bonds present in the product A is :

(A) 21 (B) 9 (C) 24 (D) 18

CHC226

81. The product formed when acetylene is passed through red hot tube is :

(A) Benzene (B) Cyclohexane (C) Neoprene (D) Ethane

CHC227

Introduction of Benzene and it's structures

82. Benzene is a resonance hybrid mainly of two Kekule's structures. Hence :

(A) Half of the molecules correspond to one structure, and half of the second structure

(B) At low temperatures benzene can be separated into two structures

(C) Two structures make equal contribution to resonance hybrid

(D) An individual benzene molecule changes back and forth between two structures

CHC236

For More Join: @IITJEE_Advanced

Chemical reactions of benzene ; Electrophilic Aromatic Substitution (EAS) of Benzene (Halogenation, Sulphonation, Nitration, Friedel Craft Alkylation, Friedel Craft Acylation)

83. Which of the following is not o, p-directing group

(A) $-NH_2$ (B) $-OH$ (C) $-X$(halogens) (D) $-CHO$

CHC239

84. In the compound C_6H_5Z which of the following is predominantly ortho/para directing?

(A) Z = $-NO_2$, —Cl, —OH (B) Z = $-OMe$, —CN, $—NH_2$

(C) Z = $-NHCOCH_3$, —Cl, —COOH (D) Z = $-NHCOCH_3$, $—CH_3$

CHC240

85. In a reaction of C_6H_5Y, the major product (>60%) is m-isomer, so the group Y is:

(A) $-COOH$ (B) $-Cl$ (C) $-OH$ (D) $-NH_2$

CHC242

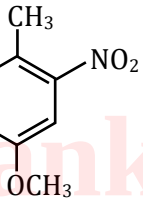
86. Reaction of SO_3 is easier in:

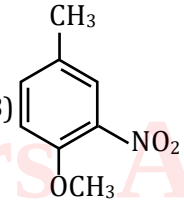
(A) Benzene (B) Toluene

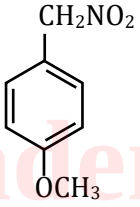
(C) Nitrobenzene (D) chlorobenzene

CHC243

87. If p-methoxy toluene is nitrated, the major product is:

(A) 

(B) 

(C) 

(D) No reaction

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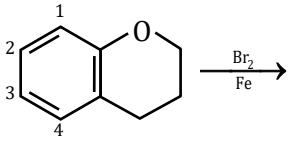
CHC246

88. Reaction of benzene with [Cl_2 + $AlCl_3$] is an example of :

(A) Electrophilic substitution reaction (B) Nucleophilic substitution reaction

(C) Electrophilic addition reaction (D) Nucleophilic addition reaction

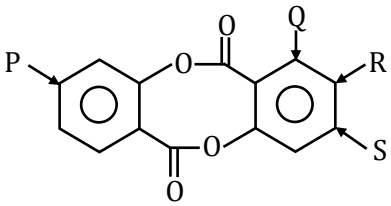
CHC247

89.  substitution take place at the position.

(A) 1 (B) 2 (C) 3 (D) both (A) & (C)

CHC248

90. The most favourable site for electrophilic substitution in the given compound is:

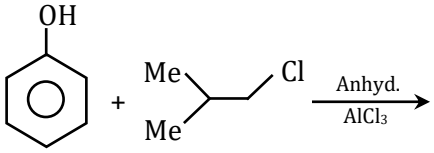


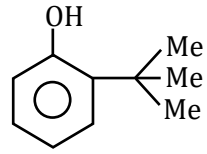
(A) P (B) Q (C) R (D) S

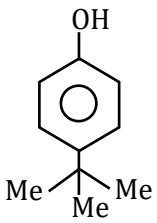
CHC252

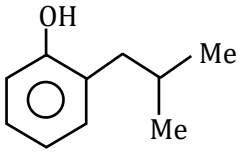
For More Join: @IITJEE_Advanced

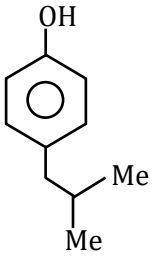
91. What will be the major product of the following reaction?



(A) 

(B) 

(C) 

(D) 

CHC253

92. Which of the following is the correct order of the rate of reaction of C_6H_6, C_6D_6 and C_6T_6 towards sulphonation?

(A) Same rates of reactions of C_6H_6, C_6D_6 and C_6T_6

(B) $C_6T_6 > C_6D_6 > C_6H_6$

(C) $C_6H_6 > C_6D_6 > C_6T_6$

(D) $C_6H_6 > C_6D_6 = C_6T_6$

CHC254

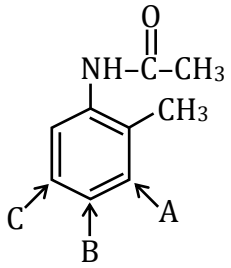
93. Which of the following is the correct order of the rate of reaction of C_6H_6, C_6D_6 and C_6T_6, towards nitration?

(A) $C_6H_6 > C_6D_6 > C_6T_6$ (B) $C_6H_6 = C_6D_6 = C_6T_6$

(C) $C_6H_6 > C_6D_6 = C_6T_6$ (D) $C_6T_6 > C_6D_6 > C_6H_6$

CHC255

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94. 

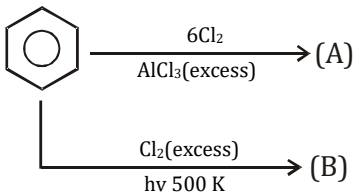
Identify the position where electrophilic aromatic substitution (EAS) is most favourable.

(A) A (B) B (C) C (D) A and C

CHC020

Chemical reaction of benzene; reaction with Cl_2

95. For the reaction



(A) A is not aromatic (B) B is aromatic

(C) A is aromatic (D) B is hexachlorobenzene

CHC058

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Preparation of alkene by Kolbe's Electrolysis

1. Aqueous solution of potassium propanoate is electrolysed. Possible organic products are:

(A) n-Butane (B) $C_2H_5COOC_2H_5$ (C) $CH_3–CH_3$ (D) $CH_2=CH_2$

CHC261

Basic idea of Elimination reactions

2. In which of the following reaction, product formed is according to Hoffmann's rule ?

(A) $CH_3CH_2—\underset{\underset{Br}{|}}{CH}—CH_3 \xrightarrow[\Delta]{t\text{-}BuO^{\ominus}K^{\oplus}}$ (B) $CH_3CH_2—\underset{\underset{Br}{|}}{CH}—CH_3 \xrightarrow[\Delta]{CH_3CH_2O^{\ominus}K^{\oplus}}$

(C) $CH_3CH_2—\underset{\underset{CH_3}{|}}{CH}—\overset{\oplus}{N}\begin{matrix}(CH_3)_2\\ \\ CH_3\end{matrix} \xrightarrow[\Delta]{OH^{\ominus}}$ (D) $CH_3CH_2\underset{\underset{\oplus S(CH_3)_2}{|}}{CH}—CH_3 \xrightarrow[\Delta]{OH^{\ominus}}$

CHC176

3. Which of the following elimination reactions will occur to give but-1-ene as the major product?

(A) $CH_3.CHCl.CH_2.CH_3 + KOH \xrightarrow{EtOH}$ (B) $CH_3.\underset{\underset{NMe_3^+}{|}}{CH}.CH_2.CH_3 + NaOEt \xrightarrow[\Delta]{EtOH}$

(C) $CH_3.CH_2.CHCl.CH_3 + Me_3COK^+ \xrightarrow{\Delta}$ (D) $CH_3.CH_2.CH(OH).CH_3 + \text{conc. } H_2SO_4 \xrightarrow[\Delta]{}$

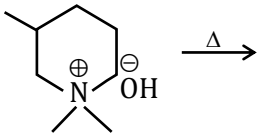
CHC262

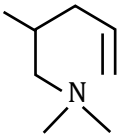
4.

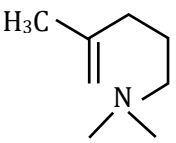
Column I	Column II
(A) $CH_3–C\equiv C–CH_3 \longrightarrow$ cis-2-butene	(P) $Na/NH_3(l)$
(B) $CH_3–C\equiv C–CH_3 \longrightarrow$ trans-2-butene	(Q) $H_2/Pd/BaSO_4$
(C) $CH_3C\equiv C–CH_3 \longrightarrow$ 1-Butyne	(R) Alc. KOH/Δ
(D) $CH_3–CH_2–C\equiv CH \longrightarrow$ 2-Butyne	(S) $NaNH_2/\Delta$

CHC263

5. Which of following are correct for given reaction



(A) Major product of reaction is 

(B) Major product is 

(C) The reaction is thermal elimination reaction

(D) The reaction is E_2 reaction

CHC061

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For More Join: @IITJEE_Advanced

Chemical reaction of alkene; basic idea of addition reaction

6. Which of the following will give same product with HBr in presence or absence of peroxide.

(A) Cyclohexene (B) 1-methylcyclohexene

(C) 1,2-dimethylcyclohexene (D) 1-butene

CHC264

Ozonolysis of alkenes (Reductive and oxidative)

7. An alkene on ozonolysis yields only ethanal. The ozonolysis of isomer of this alkene yields:

(A) Propanone (B) Ethanal

(C) Methanal (D) Only propanal

CHC265

Allylic substitution (NBS/Br_2)

8. $CH_2 = CHCH_2CH = CH_2 \xrightarrow{NBS}$ A, A can be

(A) $CH_2 = \underset{\displaystyle Br}{\underset{|}{C}}HCH = CH_2$ (B) $CH_2{=}CHCH{=}CH{-}CH_2Br$

(C) $CH_2 = CHCH_2CH = CHBr$ (D) $CH_2 = \underset{\displaystyle Br}{\underset{|}{C}}HCH_2C = CH_2$

CHC266

9. Match List-I with List-II and select the correct answer using the codes given below the lists:

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List-I (Reaction)	**List-II (Reagents)**
(A) $CH_3{-}CH{=}CH_2 \rightarrow CH_3{-}CHBr{-}CH_3$	(P) HBr
(B) $CH_3{-}CH{=}CH_2 \rightarrow CH_3{-}CH_2{-}CH_2Br$	(Q) Br_2 / CCl_4
(C) $CH_3{-}CH{=}CH_2 \rightarrow BrCH_2{-}CH{=}CH_2$	(R) HBr / Peroxide
(D) $CH_3{-}CH{=}CH_2 \rightarrow CH_3{-}CHBr{-}CH_2Br$	(S) NBS

CHC267

Preparations of Alkynes; special preparation (By Carbide, By CHX_3 + Ag)

10. CH_4 can be prepared by the reaction of H_2O with :-

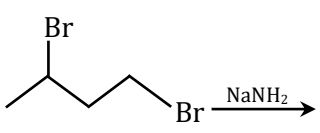
(A) Mg_2C_3 (B) CaC_2

(C) Be_2C (D) Al_4C_3

CHC234

11. In which of the following reactions(s), an alkyne product is formed?

(A) $Al_4C_3 + H_2O \longrightarrow$ (B) $CaC_2 + H_2O \longrightarrow$

(C) $Mg_2C_3 + H_2O \longrightarrow$ (D) 

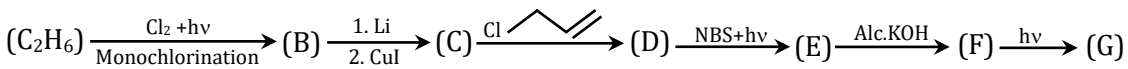
CHC235

For More Join: @IITJEE_Advanced

PARAGRAPH TYPE QUESTIONS

Basic idea of Elimination reactions

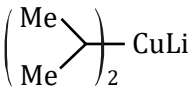
PARAGRAPH for Q. 12 to 15



12. Compound (C) is:

(A) $(C_2H_5)_2$ CuLi

(B) $(C_3H_7)_2$ CuLi

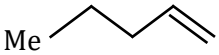
(C) 

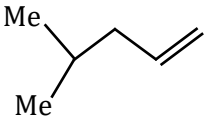
(D) 

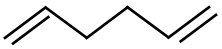
CHC063

13. Compound (D) is:

(A) 

(B) 

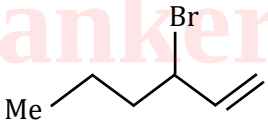
(C) 

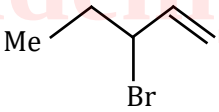
(D) 

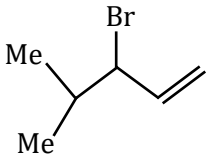
CHC064

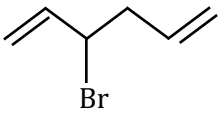
14. Compound (E) is:

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(A) 

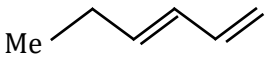
(B) 

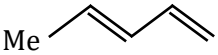
(C) 

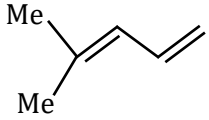
(D) 

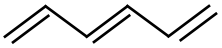
CHC065

15. Compound (F) is:

(A) 

(B) 

(C) 

(D) 

CHC066

For More Join: @IITJEE_Advanced

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Preparation of alkane by hydrogenation of alkene

1. (1) $C_2FClBrI$ (all isomers) $\xrightarrow[Ni]{H_2}$ (A) (exclude stereoisomer)

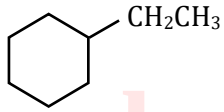
(2) C_4H_8 (alkene) (all isomers) $\xrightarrow[Ni]{H_2}$ (B) (exclude stereoisomer)

Total number of products A and B (i.e. A + B) is equal to :

CHC023

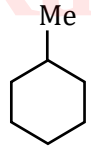
Chemical reaction of alkane; Halogenation (Free radical substitution), Reactivity and selectivity principle, Relative rate of reaction (X_2, Alkane), Reed Reaction

2. How many product are formed from the monobromination of ethylcyclohexane (excluding stereoisomers)?



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CHC128

3.  $\xrightarrow{Br_2/h\nu}$ Monobromo derivatives

The number of possible monobromo products is (excluding stereoisomers):

CHC130

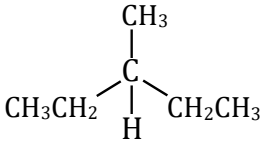
4. $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_2-\underset{\overset{|}{CH_3}}{CH}-CH_3 \xrightarrow{Br_2/h\nu} CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_2-\overset{\overset{CH_3}{|}}{\underset{\underset{Br}{|}}{C}}-CH_3$

2- Bromo-2, 4, 4-trimethyl pentane (x%)

What is the value of x (% yield of product)?

CHC026

5. The maximum number of isomers (including stereoisomers) that are possible on monochlorination of the following compound is :



CHC030

6. Number of bromo derivatives obtained on treating ethane with excess of Br_2, in diffused sunlight is...

CHC268

For More Join: @IITJEE_Advanced

Reactions of alkanes; combustion; controlled oxidation [$KMnO_4$, Cu/Heat, Mo_2O_3, $(CH_3COO)_2Mn$]; isomerization; pyrolysis; aromatization and reaction with steam

7. 10 mL of gaseous hydrocarbon on combustion gives 40 mL of $CO_2(g)$ and 50 mL of water vapour. Total number of carbon and hydrogen atoms in the hydrocarbon is ____.

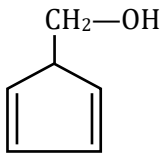
CHC269

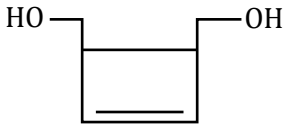
Basic idea of Elimination reactions

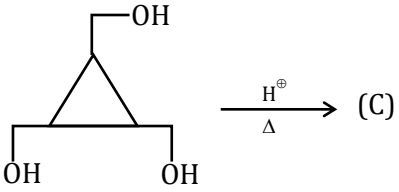
8. The total number of alkenes possible by dehydrobromination of 3-bromo-3 cyclopentylhexane using alcoholic KOH is .....

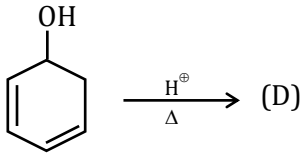
CHC178

9.

(1) 

(2) 

(3) 

(4) 

Sum of molecular mass of A, B, C, D (i.e. A + B + C + D) is equal to :

CHC022

Chemical reaction of alkene; basic idea of addition reaction

10. 3-Methylhex-2-ene on reaction with HBr in presence of peroxide forms an addition product (A). The number of possible stereoisomers for 'A' is_____.

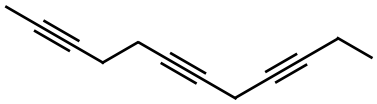
CHC270

11. The number of possible isomeric products formed when 3-chloro-1-butene reacts with HCl through carbocation formation is ____

CHC271

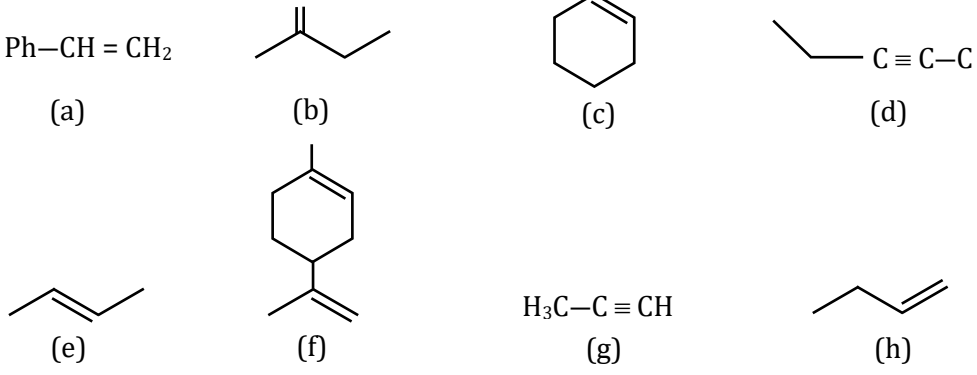
Ozonolysis of alkenes (Reductive and oxidative)

12. How many compounds are formed upon the ozonolysis of the following triyne?



CHC193

13. How many of the following compounds will produce CO_2 on oxidative ozonolysis?



(a) (b) (c) (d)

(e) (f) (g) (h)

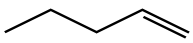
CHC194

For More Join: @IITJEE_Advanced

Allylic substitution (NBS/Br_2)

14. How many products will be formed excluding stereo when 1,3,5-trimethyl cyclohexene reacts with NBS?

CHC187

15.  Product(s).

Number of allyl bromo derivatives formed during reaction will be :

CHC192

Preparations of Alkynes; special preparation (By Carbide, By CHX_3 + Ag)

16. Calculate degree of unsaturation upon hydrolysis of Mg_2C_3.

CHC218

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

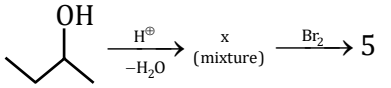
Reactions of alkanes; combustion; controlled oxidation[$KMnO_4$, Cu/Heat, Mo_2O_3, $(CH_3COO)_2Mn$]; isomerization; pyrolysis; aromatization and reaction with steam,

17. How much volume of air (in litre) will be needed for complete combustion of 10 lit. of ethane. (Assuming that approx. 20% O_2 is present in air)

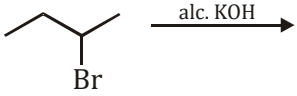
CHC068

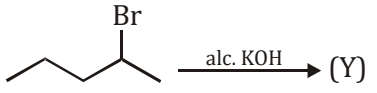
Rankers Academy JEE

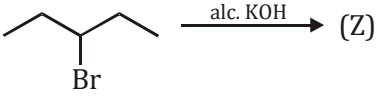
Basic idea of Elimination reactions

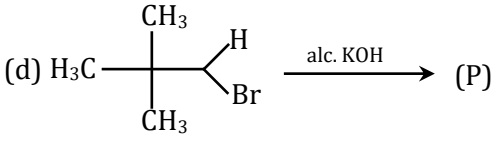
18.  5 (products) compounds of molecular formula $C_4H_8Br_2$ Number of compounds in X will be:

CHC069

19. (a)  (X)

(b)  (Y)

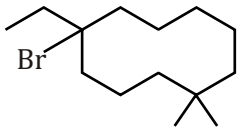
(c)  (Z)

(d)  (P)

Find the sum of elimination products including stereoisomers (X + Y + Z + P)?

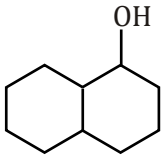
CHC196

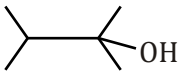
For More Join: @IITJEE_Advanced

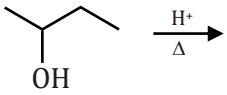
20. 

Number of possible alkenes formed in product mixture

CHC272

21. Reaction -I : 

Reaction -II : 

Reaction -III : 

Sum of number of α-H in major products (A) + (B) + (C)

CHC273

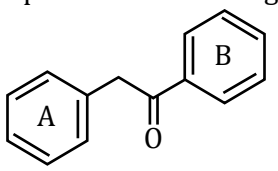
Chemical reaction of alkene; basic idea of addition reaction

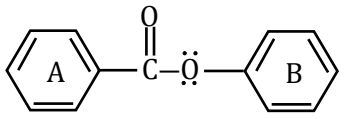
22. Number of possible compounds of molecular formula C_5H_{10} which can decolourise bromine water :

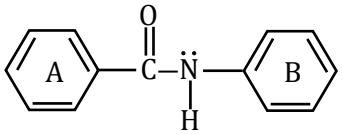
CHC274

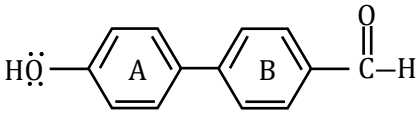
Chemical reactions of benzene ; Electrophilic Aromatic Substitution (EAS) of Benzene (Halogenation, Sulphonation, Nitration, Friedel Craft Alkylation, Friedel Craft Acylation)

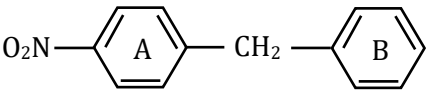
23. Each of the compounds shown below has two aromatic rings, labelled as A and B. Identify number of compounds in which ring B is more active than ring A for electrophilic aromatic substitution reaction.

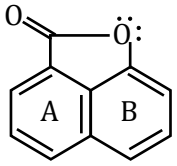
(i) 

(ii) 

(iii) 

(iv) 

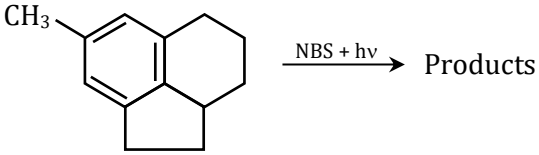
(v) 

(vi) 

CHC024

Allylic substitution (NBS/Br_2)

24. Identify the number of products (excluding sterioisomers) in the following reaction.



CHC029

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. The hydrocarbon (X) with molar mass 80 g mol^{-1} and 90% carbon has _____ degree of unsaturation.

[JEE (Main) 2025]

CHC316

2. Number of alkanes obtained on electrolysis of a mixture of CH_3COONa and C_2H_5COONa is____.

[JEE (Main) 2024]

CHC275

3. Maximum number of isomeric monochloro derivatives which can be obtained from 2, 2, 5, 5- tetramethylhexane by chlorination is _____.

[JEE (Main) 2023]

CHC276

4. Which hydrogen in compound (E) is easily replaceable during bromination reaction in presence of light :

$\underset{\delta}{CH_3}-\underset{\gamma}{CH_2}-\underset{\beta}{CH}=\underset{\alpha}{CH}$ **[JEE (Main) 2019]**

(1) β – hydrogen (2) γ – hydrogen (3) δ – hydrogen (4) α – hydrogen

CHC032

5. Number of moles of methane required to produce 22g $CO_{2(g)}$ after combustion is x × 10^{-2} moles. The value of x is **[JEE (Main) 2024]**

CHC277

6. The number of moles of methane required to produce 11g $CO_2(g)$ after complete combustion is: (Given molar mass of methane in g mol^{-1} : 16) **[JEE (Main) 2024]**

(1) 0.75 (2) 0.25 (3) 0.35 (4) 0.5

CHC278

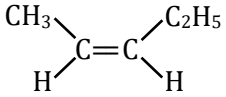
7. Which of the following is Lindlar catalyst? **[JEE (Main) 2021]**

(1) Zinc chloride and HCl (2) Cold dilute solution of $KMnO_4$

(3) Sodium and Liquid NH_3 (4) Partially deactivated palladised charcoal

CHC048

8. In the given reactions identify A and B. **[JEE (Main) 2024]**

$H_2 + A \xrightarrow{Pd/C}$ 

$CH_3 - C \equiv C - CH_3 + H_2 \xrightarrow{Na/Liquid\,NH_3} "B"$

(1) A : 2–Pentyne B : trans – 2 – butene (2) A : n – Pentane B : trans – 2 – butene

(3) A : 2 – Pentyne B : Cis – 2 – butene (4) A : n – Pentane B : Cis – 2 – butene

CHC279

9. The one giving maximum number of isomeric alkenes on dehydrohalogenation reaction is (excluding rearrangement) **[JEE (Main) 2023]**

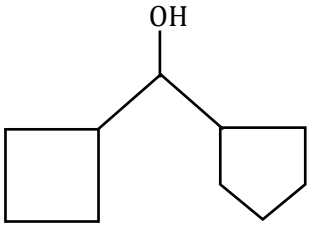
(1) 1-Bromo-2-methylbutane (2) 2-Bromopropane

(3) 2-Bromopentane (4) 2-Bromo-3,3-dimethylpentane

CHC280

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10. 

$\xrightarrow[\Delta]{H^+}$ Major product **[JEE (Main) 2023]**

In the above reaction. Left hand side and right hand side rings are named as 'A' and 'B' respectively. They undergo ring expansion. The correct statement for this process is:

(1) Finally, both rings will become six membered each.

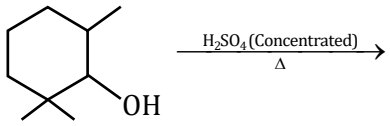
(2) Finally, both rings will become five membered each.

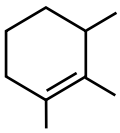
(3) Only 'A' will become 6 membered.

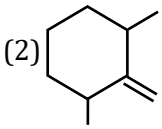
(4) Ring expansion can go upto seven membered rings

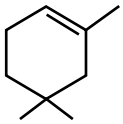
CHC281

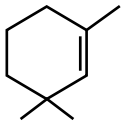
11. Find out the major product from the following reaction. **[JEE (Main) 2023]**



(1) 

(2) 

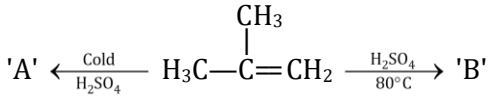
(3) 

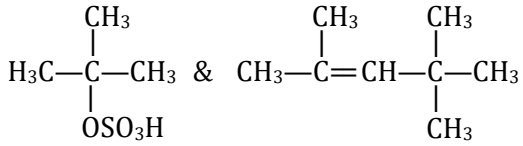
(4) 

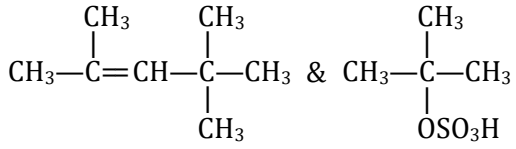
CHC258

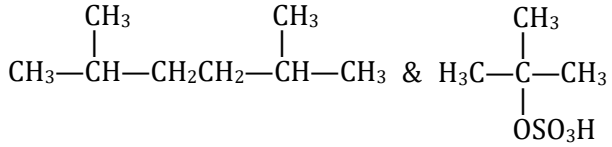
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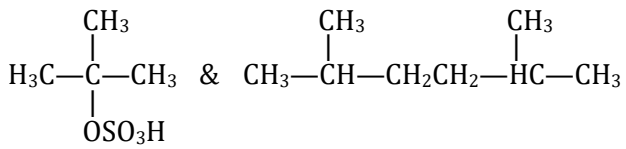
12. The major products 'A' and 'B', respectively, are **[JEE (Main) 2023]**



(1) 

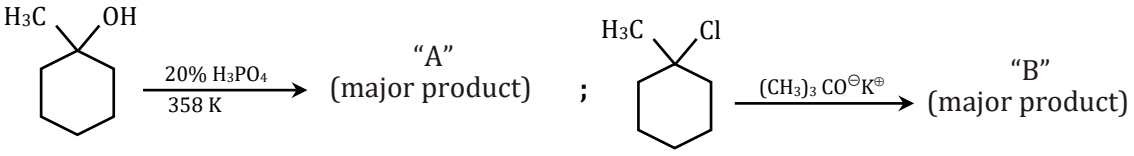
(2) 

(3) 

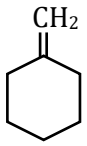
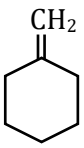
(4) 

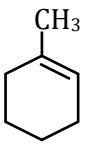
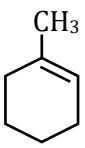
CHC259

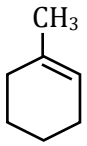
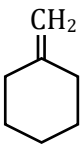
For More Join: @IITJEE_Advanced

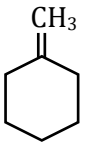
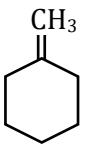
13. 

The product "A" and "B" formed in above reactions are: **[JEE (Main) 2021]**

(1) A-  B- 

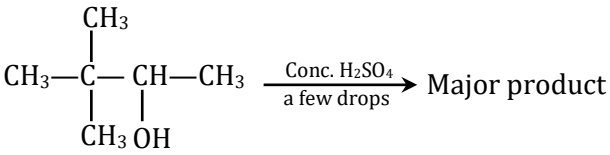
(2) A-  B- 

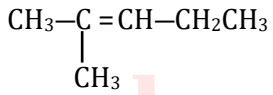
(3) A-  B- 

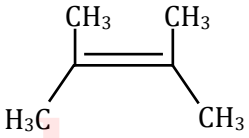
(4) A-  B- 

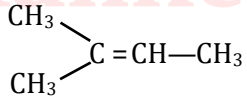
CHC049

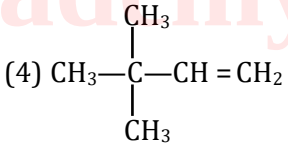
14. The major product formed in the following reaction is: **[JEE (Main) 2021]**



(1) 

(2) 

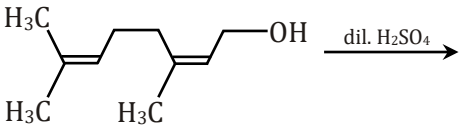
(3) 

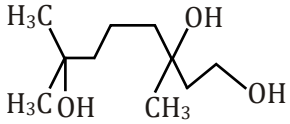
(4) 

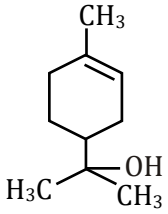
Rankers Academy JEE

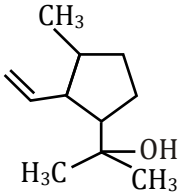
CHC051

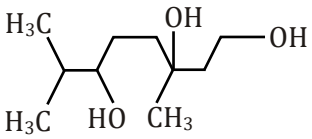
15. The major product of the following reaction is: **[JEE (Main) 2020]**



(1) 

(2) 

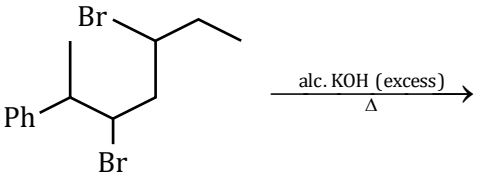
(3) 

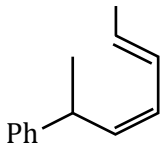
(4) 

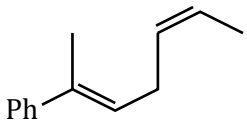
CHC040

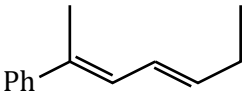
For More Join: @IITJEE_Advanced

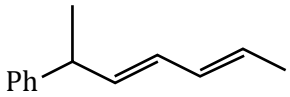
16. The major product of the following reaction is **[JEE (Main) 2019]**



(1) 

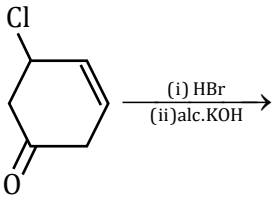
(2) 

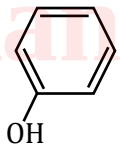
(3) 

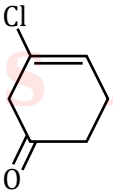
(4) 

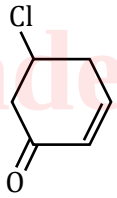
CHC031

17. The major product of the following reaction is: **[JEE (Main) 2019]**



(1) 

(2) 

(3) 

(4) 

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CHC033

18. The incorrect statement regarding the geometrical isomers of 2-butene is:

[JEE (Main) 2024]

(1) cis-2-butene and trans-2-butene are not interconvertible at room temperature.

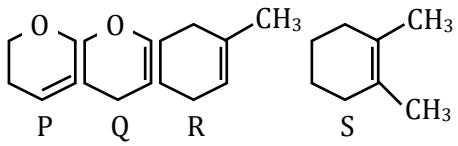
(2) cis-2-butene has less dipole moment than trans-2-butene.

(3) trans-2-butene is more stable than cis-2-butene.

(4) cis-2-butene and trans-2-butene are stereoisomers.

CHC282

19. Following are the four molecules "P", "Q", "R" and "S". Which one among the four molecules will react with H-Br(aq) at the fastest rate ? **[JEE (Main) 2025]**

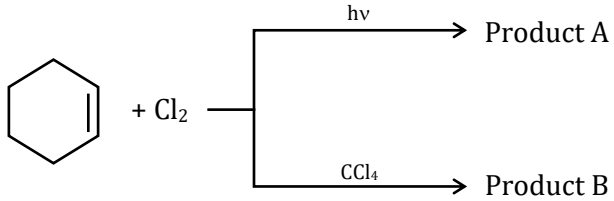


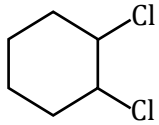
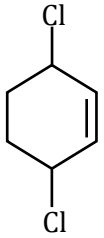
(1) S (2) Q (3) R (4) P

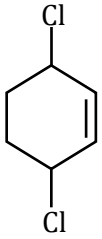
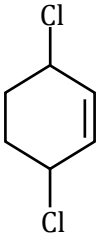
CHC317

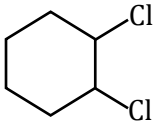
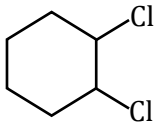
For More Join: @IITJEE_Advanced

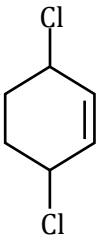
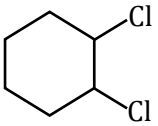
20. Identify product A and product B : **[JEE (Main) 2024]**



(1) A :  B : 

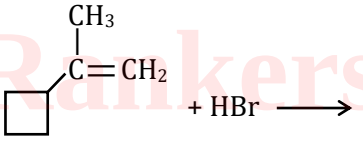
(2) A :  B : 

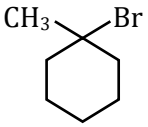
(3) A :  B : 

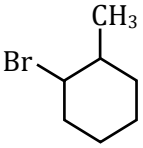
(4) A :  B : 

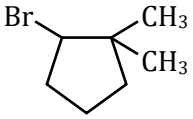
CHC283

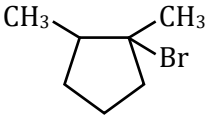
21. In the following given reaction 'A' is **[JEE (Main) 2023]**



(1) 

(2) 

(3) 

(4) 

CHC284

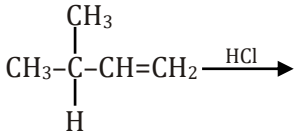
22. $2-\text{hexene} \xrightarrow[(ii)H_2O]{(i)O_3} \text{Products}$ **[JEE (Main) 2023]**

The two products formed in above reaction are -

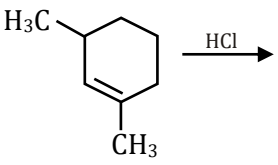
(1) Butanoic acid and acetic acid
(2) Butanal and acetic acid
(3) Butanal and acetaldehyde
(4) Butanoic acid and acetaldehyde

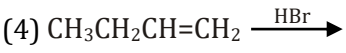
CHC285

23. Which of the following reactions will not produce a racemic product ? **[JEE (Main) 2020]**

(1) 

(2) 

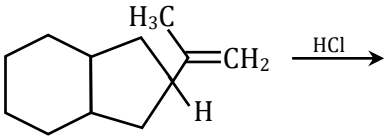
(3) 

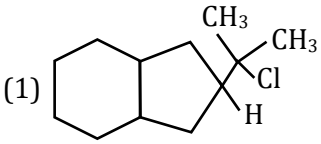
(4) 

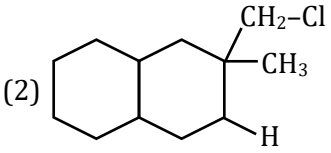
CHC043

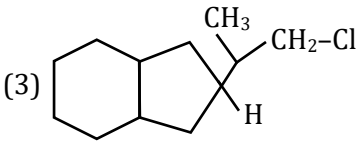
For More Join: @IITJEE_Advanced

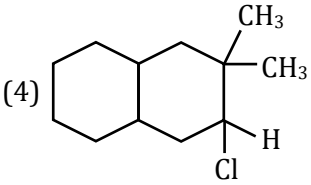
24. The major product of the following reaction is: **[JEE (Main) 2019]**



(1) 

(2) 

(3) 

(4) 

CHC034

25. Identify product [A], [B] and [C] in the following reaction sequence : **[JEE (Main) 2025]**

$CH_3-C\equiv CH \xrightarrow[H_2]{Pd/C} [A] \xrightarrow[(ii)\,Zn,H_2O]{(i)\,O_3} [B]+[C]$

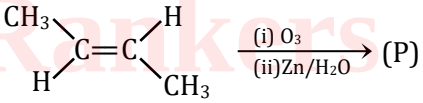
(1) [A] : CH_3–CH=CH_2, [B] : CH_3CHO, [C] : HCHO

(2) [A] : CH_2=CH_2, [B] : $H_3C-\overset{\overset{O}{\|}}{C}-CH_3$ [C] : HCHO

(3) [A] : CH_3–CH=CH_2, [B] : CH_3CHO, [C] : CH_3CH_2OH

(4) [A] : $CH_3CH_2CH_3$, [B] : CH_3CHO, [C] : HCHO

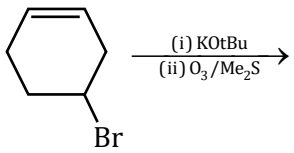
CHC318

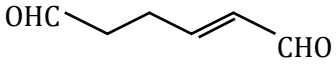
26. 

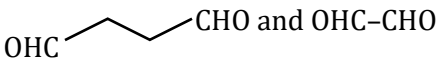
Consider the given reaction. The total number of oxygen atoms present per molecule of the product (P) is___. **[JEE (Main) 2024]**

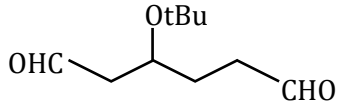
CHC286

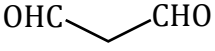
27. The major product(s) obtained in the following reaction is/are : **[JEE (Main) 2019]**



(1) 

(2) 

(3) 

(4) 

CHC036

28. The major product of the following reaction is P.

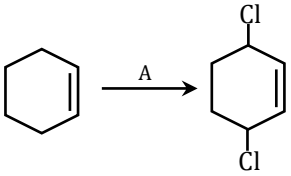
$CH_3C\equiv C-CH_3 \xrightarrow[(ii)\ dil.KMnO_4,\ 273K]{(i)\ Na/liq.NH_3}$ 'P'

Number of oxygen atoms present in product 'P' is _____ (nearest integer).

[JEE (Main) 2024]

CHC287

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29. 

Identify the reagent(s) 'A' and condition(s) for the reaction : **[JEE (Main) 2021]**

(1) A = HCl; Anhydrous $AlCl_3$ (2) A = HCl, $ZnCl_2$

(3) A = Cl_2; UV light (4) A = Cl_2; dark, Anhydrous $AlCl_3$

CHC050

30. The major product of the following reaction is: **[JEE (Main) 2021]**

$CH_3CH_2CH=CH_2 \xrightarrow[Rh\ catalyst]{H_2/CO}$

(1) $CH_3CH_2CH=CH-CHO$ (2) $CH_3CH_2\underset{\underset{CHO}{|}}{C}=CH_2$ (3) $CH_3CH_2CH_2CH_2CHO$ (4) $CH_3CH_2CH_2CHO$

CHC046

31. The major product of the following reaction is: **[JEE (Main) 2019]**

$CH_3CH_2\underset{\underset{Br}{|}}{CH}-\underset{\underset{Br}{|}}{CH_2} \xrightarrow[(ii)\ NaNH_2\ in\ liq\ NH_3]{(i)\ alc.KOH}$

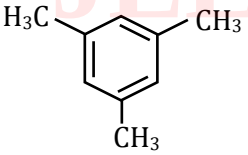
(1) $CH_3CH_2C\equiv CH$ (2) $CH_3CH_2\underset{\underset{NH_2}{|}}{CH}-\underset{\underset{NH_2}{|}}{CH_2}$ (3) $CH_3CH=C=CH_2$ (4) $CH_3CH=CHCH_2NH_2$

CHC035

32. For the given reaction : **[JEE (Main) 2021]**

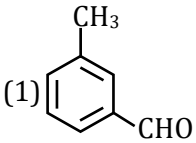
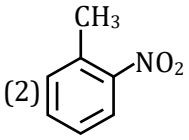
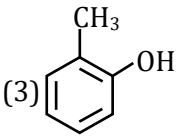
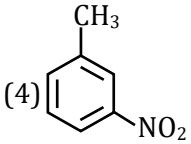
$HC(CH_3)=CHBr \xrightarrow[2.\ Red\ hot\ iron\ tube,\ 873\ K]{1.\ NaNH_2}$ (A) (major product)

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(1) $CH_3CH_2CH_2NH_2$ (2) $CH(CH_3)=CH-NH_2$ (3)  (4) 

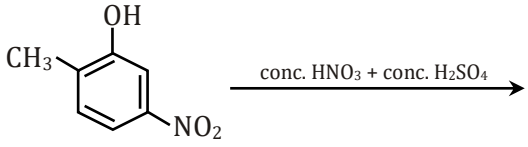
CHC047

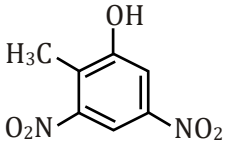
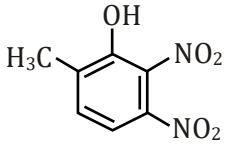
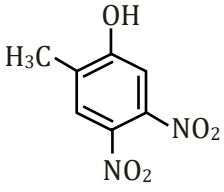
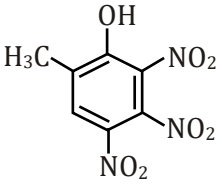
33. Halogenation of which one of the following will yield m-substituted product with respect to methyl group as a major product? **[JEE (Main) 2022]**

(1)  (2)  (3)  (4) 

CHC257

34. The major product of the following reaction is: **[JEE (Main) 2020]**

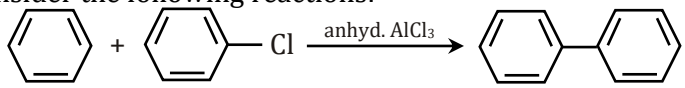


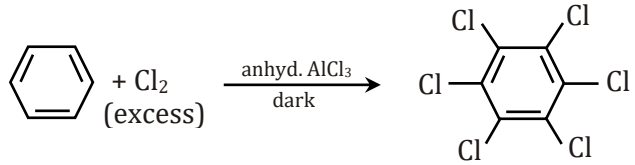
(1)  (2)  (3)  (4) 

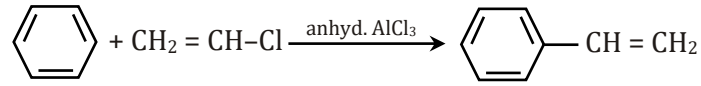
CHC044

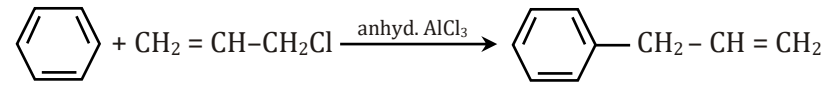
For More Join: @IITJEE_Advanced

35. Consider the following reactions: **[JEE (Main) 2020]**

(a) 

(b) 

(c) 

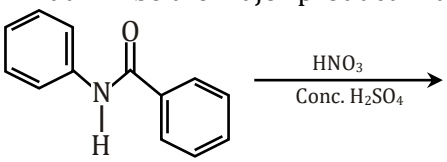
(d) 

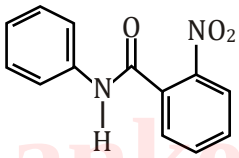
Which of these reactions are possible ?

(1) (a) and (d) (2) (b) and (d) (3) (a) and (b) (4) (b), (c) and (d)

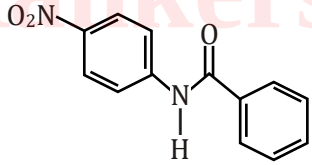
CHC045

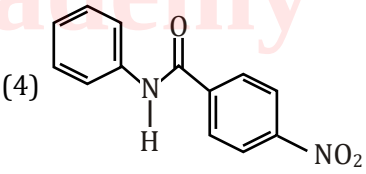
36. What will be the major product in the following mononitration reaction ? **[JEE (Main) 2019]**



(1) 

(2) 

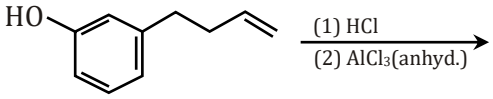
(3) 

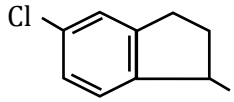
(4) 

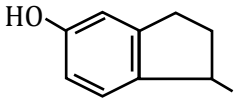
Rankers Academy JEE

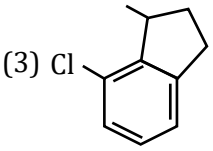
CHC037

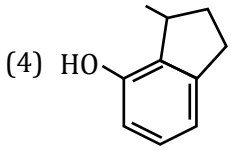
37. The major product of the following reaction is :- **[JEE (Main) 2019]**



(1) 

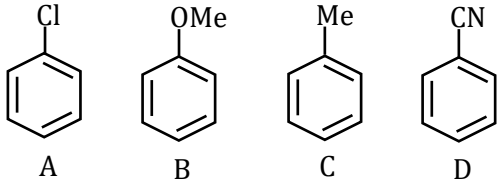
(2) 

(3) 

(4) 

CHC038

38. The increasing order of reactivity of the following compounds towards aromatic electrophilic substitution reaction is : **[JEE (Main) 2019]**



(1) D < B < A < C (2) A < B < C < D (3) D < A < C < B (4) B < C < A < D

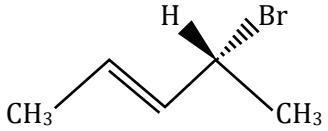
CHC039

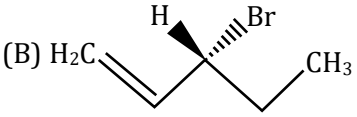
For More Join: @IITJEE_Advanced

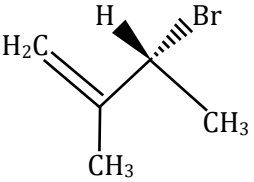
EXERCISE - JEE (Advanced) PYQ

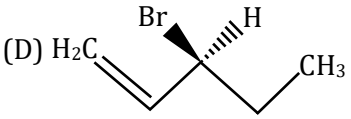
1. Compound(s) that on hydrogenation produce(s) optically inactive compound(s) is (are)

[JEE (Advanced) 2015]

(A) 

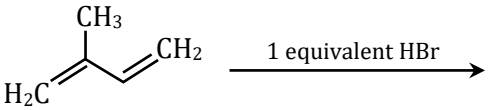
(B) 

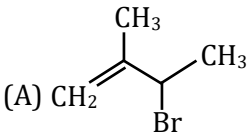
(C) 

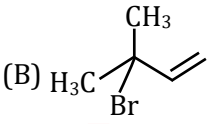
(D) 

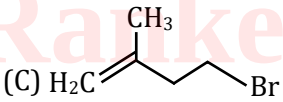
CHC075

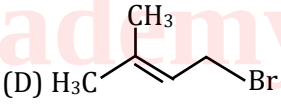
2. In the following reaction, the major product is - **[JEE (Advanced) 2015]**



(A) 

(B) 

(C) 

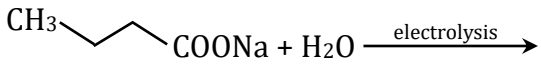
(D) 

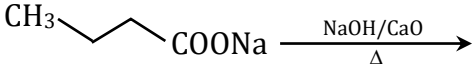
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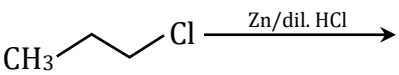
CHC078

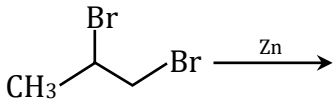
3. Which of the following reactions produce(s) propane as a major product?

[JEE (Advanced) 2019]

(A) 

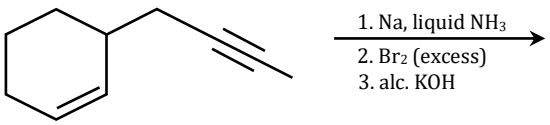
(B) 

(C) 

(D) 

CHC081

4. The number of isomeric tetraenes (**NOT** containing *sp*-hybridized carbon atoms) that can be formed from the following reaction sequence is _____. **[JEE (Advanced) 2022]**



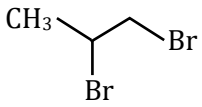
CHC085

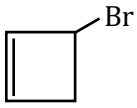
For More Join: @IITJEE_Advanced

5. Choose the correct option(s) that give(s) an aromatic compound as the major product.

[JEE (Advanced) 2019]

(A)  + Cl_2 (excess) $\xrightarrow{UV,\ 500K}$

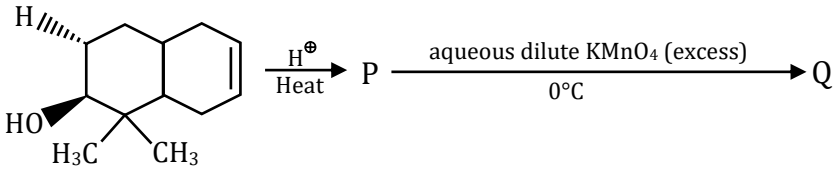
(B)  $\xrightarrow[3.\text{Redhot iron tube, 873K}]{1.\text{alcKOH}\ 2.NaNH_2}$

(C)  $\xrightarrow{NaOEt}$

(D)  $\xrightarrow{NaOMe}$

CHC082

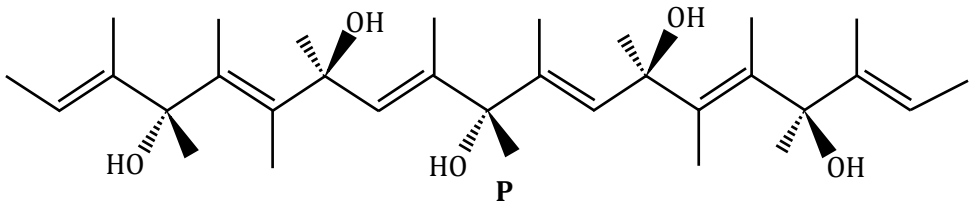
6. The number of hydroxyl group(s) in **Q** is **[JEE (Advanced) 2015]**



CHC079

7. The total number of chiral molecules formed from one molecule of **P** on complete ozonolysis (O_3, Zn/H_2O) is _____.

[JEE (Advanced) 2022]



P

CHC086

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For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	B	B	B	D	D	D	C	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	A	C	C	A	C	D	C	A	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	B	C	D	D	D	D	A	A	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	B	A	A	C	A	A	B	B	A
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	D	A	C	B	A	B	A	D	C	B
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	B	B	A	C	B	D	B	A	A	D
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	C	D	A	B	C	A	D	A	C	D
Que.	71	72	73	74	75	76	77	78	79	80
Ans.	C	B	A	D	A	B	C	C	C	A
Que.	81	82	83	84	85	86	87	88	89	90
Ans.	A	C	D	D	A	B	B	A	C	C
Que.	91	92	93	94	95					
Ans.	B	C	B	B	C					

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EXERCISE (O-2)

Que.	1	2	3	4	5	6	7
Ans.	ABCD	ACD	BC	(A)→Q ; (B)→P ; (C)→S ; (D)→R	ACD	AC	ABC
Que.	8	9		10	11	12	13	14
Ans.	AB	(A)→P ; (B)→R ; (C)→S ; (D)→Q		CD	BC	A	B	B
Que.	15							
Ans.	B							

EXERCISE (S)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	5	6	5	90	8	9	14	5	312	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	4	5	4	4	2	175.00	3.00	8.00	6.00
Que.	21	22	23	24						
Ans.	26.00	11.00	4.00	4.00						

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	3	3	2	50	2	4	1	3	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	1	3	2	2	3	1	2	2	4
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	1	1	1	1	1	2	2	3	3
Que.	31	32	33	34	35	36	37	38		
Ans.	1	4	3	3	2	3	2	3		

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	
Ans.	BD	D	BC	2	BD	4	2	

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